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MILITARY SPECIFICATION

TEST SET, FUEL-QUANTITY-GAGE: CAPACITANCE TYPE

This specification is approved for use within the Aviation Systems Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a variable capacitance fuel-quantity-gage test set applicable to aircraft (fixed wing and rotary wing) employing a capacitance type (liquid) gaging system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy, and Lead Alloy
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Systems Command, 4300 Goodfellow Blvd., ATTN: AMSAV-ELSS, St. Louis, MO 63120-1798, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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W-C-596/13	Connector, Plug, Electrical, General Purpose, Hospital Grade Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz

MILITARY

MIL-C-17	Cables, Radio Frequency, Flexible and Semirigid, General Specification for
MIL-W-76	Wire and Cable, Hook-up Electrical, Insulated
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-C-3432	Cable and Wire, Electrical (Power and Control); Semi-Flexible, (300 and 600 Volts)
MIL-C-3767	Connectors, Plug and Receptacle, Electrical, (Power, Bladed Type), General Specification for
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-G-7817	Gage, Fuel Quantity, Capacitor Type, Uncompensated, General Specification for
MIL-G-7818	Gages, Fuel Quantity, Capacitor Type, Compensated, General Specification for
MIL-G-7940	Gages, Liquid Quantity, Capacitor Type, Installation and Calibration of
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-G-8798	Gage System, Fuel-Quantity, Capacitor-Type, Non-Vacuum
MIL-S-19500	Semiconductor Devices, General Specification for

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MIL-T-28800	Test Equipment for Use With Electronic and Electrical Equipment, General Specification for
MIL-C-26482	Connector, Electric, Circular Miniature, Quick Disconnect, Environment Resisting
MIL-G-26988	Gage, Liquid Quantity, Capacitor Type, Transistorized, General Specification for
MIL-M-38510	Microcircuits, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MS3057	Clamp, Cable, Electrical Connector
MS3101	Connector, Plug, Electric Cable Connecting
MS3102	Connector, Receptacle, Electric, Box Mounting
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric, Angle 90°
MS3116	Connector, Plug, Electric, Solder Type, Straight, Bayonet Coupling

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MS3126 Connector, Plug, Electric, Crimp Type, Straight, Bayonet Coupling

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

467-000-P00(FSCM 26055) Cable Assemblies, Army Test Set

NATIONAL BUREAU OF STANDARDS

Certificate of Calibration

(Copies of Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Packaging, Commerical

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design Characteristics. The test set design shall provide a means to determine the serviceability of aircraft employing a capacitance-type fuel gaging system. It shall consist of a single integral unit incorporating a transit case, direct current (dc) electrical resistance measuring circuitry, capacitance measuring circuitry, capacitance substitution circuitry, cali-

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bration circuitry with built-in standards of capacitance and resistance, switches, wiring, cable assemblies, and accessories. The structure and feature relating to calibration shall provide for lowest cost, simplicity, accuracy, and reasonable convenience in use.

3.1.1 First Article. If first article approval and functional tests are required (see 6.2), approval and testing shall be in accordance with 4.2.1. Approval of the first article shall not relieve the contractor of his responsibility to furnish a product in accordance with this specification. Any change or deviation from the approved first article shall be subject to the prior written approval of the contracting officer.

3.2 Materials. Materials shall be as specified for the specific components; unless otherwise specified, materials shall be selected in accordance with MIL-STD-143.

3.2.1 Metals. Metals shall be of the corrosion-resistance type, unless suitably protected to resist corrosion during storage and normal service use. Unless dissimilar metals are suitably protected against electrolytic corrosion, they shall not be used in intimate contact with each other. Requirement 16 of MIL-STD-454 shall apply.

3.2.2 Nonmagnetic. Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.2.3 Fungusproof. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where nutrient materials must be used, they shall be treated with a fungicidal agent in accordance with MIL-T-152. Requirement 4 of MIL-STD-454 shall apply.

3.2.4 Environmental. The test set shall withstand the extreme conditions of temperature, vibration, shock and humidity (see 4.3.6 through 4.3.11).

3.3 Construction. The general layout of the test set and construction shall be mechanically and electrically sound, suitable for the operational requirements specified herein. The design layout, and assembly of the test set, and its component parts, shall be such as to facilitate quantity production and ease of maintenance and operation. Unless otherwise specified, design and construction shall be in accordance with MIL-S-8512.

3.3.1 Dimensions and tolerances. Unless otherwise specified herein, dimensions and tolerances shall be as close as consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability of parts, operation, or performance of the test set, they shall be held or limited accordingly.

3.3.2 Standard parts. Standard parts (AN or MS) shall be used wherever they are suitable for the purpose, and shall be identified by their part

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numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by military standard parts without alteration.

3.3.3 Threaded parts. Unless otherwise specified, machine screws, 0.060-inch (1.524 mm) or larger in diameter, shall conform to MIL-S-7742.

3.3.4 Soldering. Soldering shall be in accordance with MIL-S-6872. Solder for the electrical connections shall be of a suitable composition and in accordance with QQ-S-571. In the assembly of the electrical wiring and connector cables, only rosin, rosin and alcohol, or equivalent plastic rosin mixtures shall be use.

3.3.5 Screw assemblies. Assembly screws and bolts shall be tight. "Tight" means without damage to the screw, bolt, or any other component of the assembly.

3.3.6 Protective treatments. If materials used in the construction of the test set are subject to deterioration when exposed to climatic and environmental conditions liable to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance of the test set. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Finishes and protective coatings.

3.3.7.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that dials, small holes, and case inserts need not be anodized.

3.3.7.2 Chemical film. Coating with a chemical film conforming to MIL-C-5541 may be used instead of anodizing when:

(a) Primary purpose of the treatment is to afford a suitable paint base.

(b) Aluminum alloys do not anodize satisfactorily.

3.3.7.3 Abrasion resistance. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 may not be used instead of anodizing.

3.3.7.4 Bonded aluminum surfaces. Aluminum surfaces which require bonding shall be provided with a means to insure low-resistance electrical path.

3.3.8 Interchangeability of parts. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

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3.4 Capacitance measuring section. The test set shall have a capacitance measuring section to determine the serviceability of aircraft capacitance type fuel quantity gaging systems. This section shall incorporate all of the components to produce an automatic display of the measured capacitance. The display shall be lighted digital.

3.5 Capacitance substitution section and calibrating section. The test set shall have a capacitance substitute section and calibrating section. The circuitry design incorporated in the test set shall measure and calibrate fuel quantity gage-tank units of the capacitance type conforming to MIL-G-7817, MIL-G-7818, MIL-G-7940, MIL-G-8798, and MIL-G-26988. When the test leads or adapter leads of the test set are shorted, no damage shall occur to the test set or personnel.

3.5.1 Capacitance substitution section. The capacitance substitution section shall have a minimum of three variable capacitance sections and one fixed capacitor of 2000 Pf±5%. The sections shall be electronically isolated from each other. The sections shall be brought out to separate panel connectors of the BNC type or equivalent. One of the variable sections shall, as a minimum range spread, have a minimum range spread of 15 to 8000 Pf. The other two shall, as a minimum range spread, have a minimum range spread of 15 to 375 Pf.

3.5.2 Calibration. Provisions shall be made to precisely calibrate the substitute capacitance values internally by means of the capacitance bridge. The necessary cables, shielded and unshielded, shall be provided with the test set.

3.6 Measuring ranges. The capacitance measuring section ranges shall be suitably arranged to provide readability consistent with the specified accuracy. The range of indicated electrical resistance shall cover the range from .5 megohm to 10,000 megohms.

3.7 Controls. All controls shall be readily accessible, suitably arranged, and of size and construction to permit convenience and ease of operation by operating personnel, under all service conditions.

3.8 Indicators. The indicator provided for measurement of capacitance and resistance values shall provide an in-line numeric digital display. The indicator shall be visible in bright sunlight and its operation principle shall be entirely consistent with the environmental extremes specified in Section 4. The same indicator shall be use to indicate capacitance or resistance in response to a panel mounted switch.

3.9 Electrical components.

3.9.1 Electronic units. Insofar as practicable, the electronic units for the capacitance and resistance circuits shall be modular designed and

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physically and electrically independent of one another. The calibrating capabilities shall utilize precision capacitor and resistor standards separate from those used in measuring or substitution applications.

3.9.2 Capacitance measuring section. Calibration adjustments shall be provided to permit precise settings at the zero and maximum ends of the scale. The circuit shall be designed to provide a high degree of stability to preclude the need for frequent recalibration. Sufficient adjustment range shall be provided to allow for calibration during the entire life of the instrument.

3.9.3 Resistance section. The electronic unit for the resistance-measuring circuit shall incorporate a sensitive and stable electronic, resistance measuring circuit. The dc voltage applied by the circuit across the electrodes and conductors, or both, undergoing test, shall not exceed 50 volts. Easily detachable protective covers incorporating adequate sealing provisions shall be provided on the panel for enclosing the adjustments for the capacitance-and-resistance-measuring circuits. The design shall not permit removal of the covers from the panel during normal use.

3.9.4 Explosion hazard. The capacitance-and-resistance-measuring circuits shall be designed so that the current in the adapter leads does not exceed 0.010 ampere when any combination of circuits are operating under normal conditions and the leads are shorted together, or when the circuits are abnormally operated, such as by setting the adjustments to any desired position or by shorting any capacitor or combination of capacitor or combination of capacitors in the measuring circuits. All grounds shall be isolated from input power, to prevent occurrence of a spark while test set is being used. There shall be positive continuity (zero ohms) between ground jack on tester panel and tester case, to preclude any direct short through the power cable. The test set shall be devoid of fire and explosion hazards, and completely safe in use around aircraft fuel.

3.9.4.1 Safety. Safety consideration for the test set shall be in accordance with MIL-STD-454, Requirement 1.

3.9.5 Input voltage. The test set shall be designed to operate from a power source of 47HZ to 440 HZ, 105 to 125 volts, single phase. Isolation transformers shall be incorporated in the input circuit of the test set.

3.9.6 Standard electronic parts. Electronic parts, and the application thereof, shall be in accordance with MIL-T-28800.

3.9.7 Semiconductors and microcircuits. Selection of semiconductors shall be in accordance with MIL-STD-701 and MIL-S-19500. Selection of microcircuits shall be in accordance with MIL-STD-883, Class B, Method 5004, and MIL-M-38510.

3.9.8 Capacitors. Fixed capacitors in the capacitance-measuring circuit and capacitance-substitution section of the test set shall be precision built,

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accurately calibrated, and aged to insure a high degree of stability over extended periods of use. The fixed capacitors shall be rated at a minimum of 500 volts, peak, across the plates, or to ground. The dissipation factor of the reference capacitors shall not exceed 0.001 at a frequency of 400 Hz and at a mean temperature of 25 °C. The capacitors shall be adequately sealed and meet the tests specified in Section 4. Precision variable air capacitors shall be constructed to withstand rugged and continuous use, shall provide a high degree of accuracy and stability during repeated operations, and shall meet the tests specified in Section 4. They shall demonstrate a linear capacitance characteristic.

3.9.9 Switches. Switches may be rotary or toggle constructed to withstand repeated use and suitable for their intended use. Indexing and position retention shall be positive. Size, shape, and location of switches shall in no way impair operation of the test set. In no way shall switches impair the designated functions of the test set.

3.9.10 Internal wiring. All electrical connections within the instrument case shall be supported to alleviate breakage and minimize changes in the measuring circuitry caused by strains, jars, vibration, and other conditions incident to shipping, storage, and service. Hookup wire shall conform to MIL-W-76. Soldering lugs or terminals shall be provided for all points of connection. Before soldering, all wires shall be securely fastened by crimping the wire upon the terminals or lugs. Shielded wire shall be used to minimize any stray voltage.

3.9.11 Electrical connectors. Connectors shall conform to MIL-C-3767, MIL-C-5015, or MIL-C-26482 where applicable. An electrical connector receptacle conforming to MS3102A-10SL-3P shall be located on the control panel to permit electrical connection to the input electrical power cable.

3.9.12 Electrical cable assemblies. Drawings 467-000-P00 (FSCM 26055), sheets 1, 2, 3, and 4 are provided for informational purposes to aid in cable design as necessary.

NOTE: The drawings are not intended to restrict design ingenuity, but aid in interface of the test set and the aircraft fuel quantity measuring systems. It is the responsibility of the contractor to obtain up-to-date drawings and data needed to design the cable assemblies (See 3.3.2).

3.9.12.1 Input power cable. Input power cable shall be a flexible, medium duty cable, type CO-03MGF(3/18)0340 conforming to MIL-C-3432. The overall length of the power cable shall be 21.0 feet (6.40 m), plus or minus 6 inches (15.24 cm) One end of the power cable shall have an adapter conforming to MS3057A-4A and an associated 90-degree angle conductor plug conforming to MS3108E-10SL-3S, or straight type MS3106A-10SL-3S with appropriate strain relief. The other end of the power cable shall have an adapter conforming to MS3057-4A and an associated straight conductor plug conforming to MS3106-10SL-3P. Two adapter cables having an overall length of 14.0 inches (35.56 cm), plus or minus

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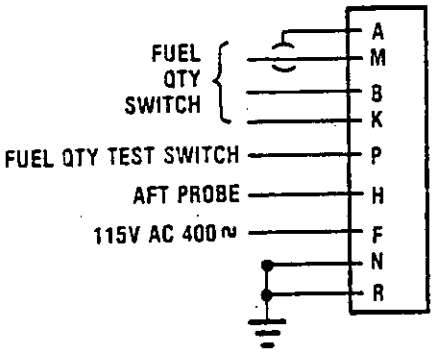
1.0 inch (2.54 cm), shall also be provided. Flexible cables of the same type specified for the power cable shall be provided for the adapter cables. One of the adapter cables shall have an adapter conforming to MS3057-4A and a cable connecting receptacle conforming to MS3101-10SL-3S on one end; on the other end, the jacket shall be removed from the cable for a length of 6.0 inches (15.24 cm), plus or minus 0.5 inch (1.27 cm). Each conductor shall have an electrical cable clip, conforming to P/N 45 (FSCM 76545) or equivalent. Each clip shall have a flexible insulator protective sleeve. At the point where the electrical conductors protrude from the jacket, means shall be provided for sealing and reinforcing the cable. The other adapter cable shall have on one end, an adapter conforming to MS3057-4A and a cable connecting receptacle conforming to MS3101-10SL-3S; the other end shall terminate with a conductor plug conforming to W-C-596/13C or equivalent.

3.9.12.2 Connector cabling. Each test set shall have a maximum of three cables with appropriate cable adapters compatible to all configurations of Army aircraft and fuel quantity measuring systems. Universal cabling and adapters shall be used, wherever practicable. These cables shall be equipped with identification tags, the material and type of marking capable of withstanding normal service usage, and the test specified in Section 4. Changeover from functional mode, such as from capacitance measuring to capacitance substitution or resistance measuring, shall not require cable removal from the test set, but must be done by internal switching. Means shall be provided, by internal switching, for the test set to act as a junction box between the aircraft tank units and the indicator when interfaced with the aircraft fuel gaging system. In addition, means shall be provided to parallel the aircraft's tank units and compensator with the test set's simulator capacitors and to supply this parallel combination to the aircraft's indicator by means of the test set's internal switching only. (See Table I for Army aircraft having capacitance type, fuel gaging system and the connectors used by these aircraft which mate with the fuel quantity indicator).

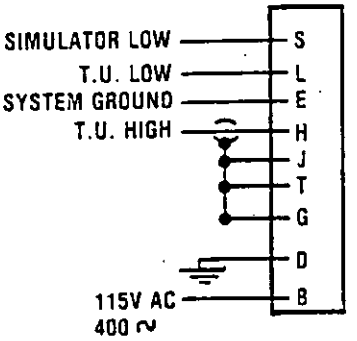
TABLE I. Army aircraft and fuel quantity indicator cable connector

Aircraft	Connector
AH-1	Amphenol Plug (02660) P/N 165-61 or Equal
UH-1	Amphenol Plug (02660) P/N 165-61 or Equal
OV-1	MS3116P-14-19S
CH-47A, B	MS3126-14-19S
CH-47C, D	MS3126-14-15S
CH-54A, B	MS3126-14-19S
CH-54B Auxiliary Tank Indicator	P/N PTO6SE14-18S (77820) or Equal

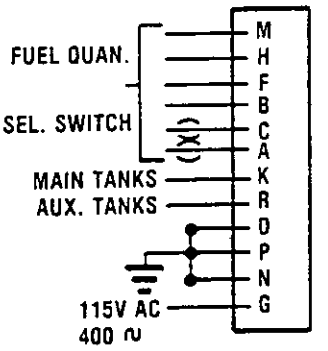
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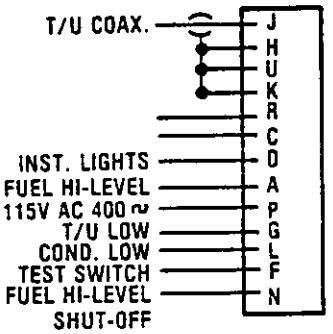
MS 3116P14-19S
OV-1



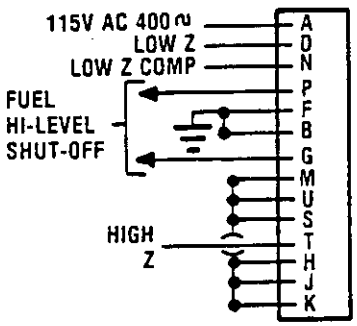
MS 3126-14-19S
CH-47A, B



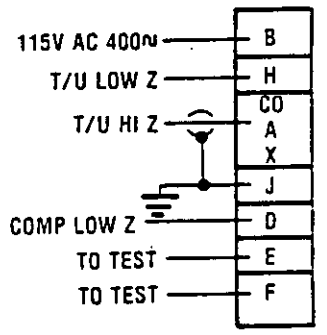
MS 3126-14-15S
CH-47C, D



CH-54A, B
FWD & AFT TANKS
MS 3126E-14-19S



CH-54B
AUX. TANK IND.
PT06SE 14-18S
OR EQUIVALENT



UH-1 & AH-1
AMPHENOL
165-61
OR EQUIVALENT

FIGURE 1. Aircraft Harness Connector Mating With Fuel Quantity Indicator

BG-213

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3.9.12.3 Test leads. A pair of heavy-duty, flexible test leads equipped with spade or banana-type terminals on one end and spring clips with protective sleeves on the other end, shall be furnished. The terminals shall mate with the binding posts for facilitating the measurement of resistance externally. The insulation for the conductors of both test leads shall be black. An additional lead of the same basic design shall be provided for grounding the test set. Length shall be determined by the contractor. (Nominal length is 15.0 feet) (4.57 m).

3.9.12.4 Adapter leads. The number of cable assemblies and adapter cables for the specific Army aircraft as listed in Table I shall also be provided. (See Figure 1 for termination of aircraft harness connector mating with fuel quantity indicator).

3.9.12.5 Coaxial cable. The coaxial cable for the connector cabling end adapter leads shall conform to type RG-58c/U of MIL-C-17 or where connector design dictates, the cables may be of smaller size but comparable quality. Stranded wire, size 20 AWG (0.032 inches) (.813 mm), for the unshielded leads shall be insulated with material entirely suitable for the purpose.

3.9.12.6 Cables. Quality of workmanship and procedures for sealing, potting, and reinforcing all cables shall conform to industry-recognized commercial standards or equivalent applicable Government Standards.

3.9.12.7 Accessories. Adequate cables, leads, and adapters shall be provided with the test set for testing all applicable aircraft.

3.10 Transit case. The transit case shall be of half hard aluminum conforming to QQ-A-250/8 minimum thickness of 0.081 inch (2.057 mm), or of material capable of meeting all requirements and tests specified herein. The instruments shall be mounted on a removable anodized aluminum panel, suitable shock protected. The transit case shall be gloss yellow, shade No. 13538 of FED-STD-595 in accordance with MIL-T-28800.

3.10.1 Dimensions. Outside dimensions of the transit case shall be a maximum of 21.0 inches (53.34 cm) by 15.0 inches (38.1 cm) by 15.0 inches (38.1 cm).

3.10.2 Handle. The transit case hinged metal handle, with grip portion of a nonmetallic material and shaped to fit the hand comfortably, shall provide a clearance from the case sufficient to pass a block 1.875 by 4.375 inches (4.76 by 11.11 cm) in cross section with edges rounded to a .9375 inch (2.38 cm) radius.

3.10.3 Lid. A detachable lid shall be secured to the transit case with a hinge. A sturdy lid support shall be fastened to the case and the lid, to prevent damage to the stowed cable and adapters and to relieve the hinge strain when the lid is open. A bracket assembly shall be provided on the underneath

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lid surface to permit the winding of ac power leads thereon for storage purposes. Means shall be provided on the inside lid surface for securing the chart of operating instructions for the test set.

3.10.4 Latches. The transit case shall have hooks and latches of corrosion resistant stainless steel, the sturdy, pull-down type for hooks on the case and lid.

3.10.5 Accessory storage. The transit case shall have compartments for storage of all accessories, excepting the input power cable. This compartment shall be at least 2.5 inches (6.35 cm) wide, and shall be free of all sharp edges, burrs, and protrusions. An optional storage compartment may be located in the lid of the test set.

3.10.6 Panel. Instrument case panel shall be fabricated of aluminum alloy conforming to QQ-A-250/8, 0.125 inch (3.175 mm) minimum thickness, or of molded phenolic 0.25-inch (6.35 mm) minimum thickness. This panel for the mounting of the indicators, electronic units, range adjustments, switches, and receptacles, and other components except the accessories and shock-mounted parts, shall be rigidly mounted to form an assembled unit. The design shall permit its ready installation and removal from the case. Its outside surface shall be black conforming to color No. 37038 in accordance with FED-STD-595.

3.10.6.1 Electronic unit mounting. The electronic units shall be shock-mounted when necessary. The shock mounts shall perform satisfactorily regardless of the position of the test set. The design shall also permit the electronic units to be readily disconnected, removed, and replaced.

3.10.6.2 Moisture pockets. All pockets, wells and traps in which water or condensed moisture can collect when the test set is in the normal operating position, shall be eliminated.

3.10.7 Sealing. The instrument case, panel, control shafts, switches, indicators, and receptacles shall be adequately sealed to meet the requirements specified herein. Gasket materials shall be fungus inert. In general, the following types of material shall be used.

a. Nylon.

b. Rubber (natural or synthetic).

3.11 Operating instructions. Operating instructions for the test set shall be attached to the inside surface of the transit case lid. Body of the chart shall be white and the markings, black. The chart shall withstand the effects of tests specified in Section 4 and normal service usage. It shall include the following:

a. Procedures to follow in operating the test set.

(Note: This data shall include the manner in which the test set shall be grounded, connected to the input power source; settings to be made prior to turning on the test set, settings to be made after turning on the test set,

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and warm-up time required; and general procedures to follow in measuring capacitance and insulation resistance of fuel gage tank units and associated connector cables. A capacitance correction chart for the use of the calibrating capability, shall be included.)

b. Illustration showing the values registered by the capacitance section indicator when the indicator is maintained at mid-scale position on the scale and with the selector switch in positions of range selection.

c. Illustration showing the values registered by the resistance section indicator when the indicator is maintained at mid-scale position and with the selector switch in positions of range selection.

3.12 Caution placard. A placard stating the voltage requirements of the test set shall be permanently and conspicuously attached to the inside of the transit case lid. An additional caution placard shall be mounted conspicuously and permanently on the panel display, indicating that the test set shall be permanently grounded prior to connection to aircraft fuel probes under test.

3.13 Identification of product. Test set, assemblies, and parts shall be identified in accordance with MIL-STD-130.

3.13.1 Transit case. Plate on the transit case shall include as a minimum, the following:

Nomenclature:	TEST SET, FUEL-QUANTITY-GAGE: CAPACITANCE TYPE
FSN:	_____
Specification:	_____
Contract No.	_____
Manufacturer's Part No.	_____

3.14 Workmanship. The test set, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges. The test set shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material during and after final assembly. Burrs and sharp edges, and resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance or the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of examinations and test. Examinations and tests of the test set shall be classified as (1) first article and (2) quality of conformance.

4.2.1 First article. If first-article approval is required (see 6.2), the first test set produced (see 3.1.1) shall be subjected to all tests specified in 4.3. If required (see 6.2), the Government will conduct functional tests of the test set and all cable adapters using the applicable aircraft shown in Table I prior to first-article approval. The functional tests will be conducted at a Government installation and at Government expense.

4.2.1.1 First article test report. A first article test report shall be submitted subsequent to the first article test. Approval of this test report shall constitute acceptance of the first article.

4.2.2 Quality conformance. Each test set shall be submitted to the tests specified in 4.3.2, 4.3.3, 4.3.4, and 4.3.5.

4.3 Test.

4.3.1 Procedures.

4.3.1.1 Position. Unless otherwise specified, the test sets shall be tested in normal operating position.

4.3.1.2 Capacitance and resistance standards. The accuracy of capacitance standards shall be 0.15 percent for capacitance of 50 pF or more, and 0.1 pF for capacitance of less than 50 pF. The sealed capacitance

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standard used for calibration of the capacitance bridge shall be accurate to 0.1 pF. Accuracy of the resistance standards employed shall be 0.1 percent, although specific values employed for values specified hereinafter may vary as much as 2 percent. The use of correction charts is permitted. The calibration accuracy of the standards shall be based on National Bureau of Standards certification.

4.3.1.3 Test set adjustment. Prior to the tests specified hereinafter, the test set shall be connected to a suitable input power source and shall be electrically energized for a 5 minute period. Immediately after the warm-up period, the capacitance and resistance sections shall be adjusted and the adjustment cover secured for the capacitance section. No further changes in the settings of the adjustments for the capacitance section shall be made during the specified tests. Unless otherwise specified, the resistance section may be readjusted prior to taking readings during the specified tests.

4.3.2 Examinations. Each test set shall be inspected to determine compliance with this specification with respect to materials, workmanship, marking, and construction.

4.3.3 Operation at room temperature. The calibration accuracy of the test set shall be determined at room temperature by means of precision capacitance and resistance standards specified in 4.3.1.2. For tests proposed standard ambient conditions are defined in MIL-STD-810.

4.3.3.1 Capacitance standard measurements. With the capacitance standard connected to receptacles appropriate on the test set by means of electrical cabling, measurements shall be conducted with the capacitance section of the test sets when the following capacitance values are introduced. The capacitance value registered by the capacitance-section indicator herein referred to as "reference values" shall not differ from the true capacitance values introduced by the capacitance standard by more than plus or minus 0.10 percent of the maximum value of each capacitance range. The values established up scale and down scale shall not differ by more than plus or minus 0.05 percent. The capacitance values specified below shall be again used to test the capacitance substitution section by means of the master capacitance bridge. The capacitance values indicated directly by the test sets shall not differ from the respective values indicated by the master capacitance bridge by more than plus or minus 0.10 percent of the maximum value of each capacitance range. The values established up scale or down scale shall not differ by more than plus or minus 0.15 percent. Values established by this test are hereinafter referred to as "reference values". Also, the capacitance of the calibration function shall be tested.

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15 pF	300 pF	3,000 pF
20 pF	375 pF	4,000 pF
40 pF	400 pF	5,000 pF
60 pF	600 pF	6,000 pF
80 pF	800 pF	7,000 pF
100 pF	1,000 pF	8,000 pF
180 pF	2,000 pF	

4.3.3.2 Resistance standard measurements. With resistance standard connected electrically across receptacles on the test sets by means of electric cabling, measurements shall be conducted with the resistance section of the test sets when the following resistance values are introduced. The resistance values registered by the resistance section indicator, herein referred to as "referenced values", shall not differ from the true resistance values introduced by the resistance standard by more than the resistance value equal to plus or minus 20 percent of true resistance values. The same measurement shall be repeated with the resistance standard electrically connected across appropriate receptacles and ground, and across the external resistance terminations.

500,000 ohms	100 megohms
1 megohm	1,000 megohms
10 megohms	10,000 megohms

4.3.3.3 Additional performance requirements. During the tests specified in 4.3.3.1, and 4.3.3.2, the following additional performance requirements shall be met:

a. The selector switches shall operate smoothly and shall be positively positioned at each respective setting.

b. The performance characteristic of the capacitance section is not adversely affected when capacitance of 10,000 pF, plus or minus 5 percent, and 10,000 pF, plus or minus 5 percent, are connected to the inner conductor of the externally shielded connector lead and ground and the conductor of the externally unshielded connector lead and ground respectively.

c. Measurements shall be made to determine that not more than 0.010 ampere, ac or dc, electrical current flows through the external cabling connected to the output receptacles when the conductors are shorted together at the outer ends of the cables, and when the capacitance and resistance measuring sections are operated over their complete measuring ranges.

4.3.4 Connector cable and adapter lead. Each connector cable and adapter lead shall be checked for continuity of wiring. In addition, the dc resistance of each cable and adapter lead shall be measured with a reliable megohmmeter. The measurements shall be taken between the center conductor and each connector shell.

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4.3.5 Voltage and frequency variation. The test set shall be checked under the combinations of frequency and voltage as specified in Table II.

TABLE II. Frequency combinations

Conditions	Voltage ± 1.0V	Frequency ± 1%
(1)	115	400 Hz
(2)	105	47 Hz
(3)	125	440 Hz

The change in indication of the capacitance and resistance section of the test set under conditions (2) and (3) from that observed under condition (1) shall not differ by more than 1 percent of the reading, or plus or minus 0.5 percent of the maximum value of each capacitance range, whichever is greater for the capacitance section, and for the resistance section, the readings shall not differ by a resistance value equal to ± 10 percent of the reading. The test shall be conducted at no less than three test points for each measuring section of the tester.

4.3.6 Low temperature. The test sets shall be placed within a test chamber with the internal temperature maintained at minus 55°C, plus or minus 2°C, for a 12-hour period. At the end of the 12-hour period, the temperature shall be raised to minus 40°C, plus or minus 2°C, for a 2-hour period. At the end of this period, and while the test sets are still at the low temperature, the test sets shall be examined and checked to determine that the calibration accuracy does not differ form the reference values by more than plus or minus 1 percent of the reading, or plus or minus 0.5 percent of the maximum value of each capacitance range, whichever is greater. For the resistance section, the readings shall be noted and shall not differ from the reference values by a resistance value equal to plus or minus 10 percent of the reference values. Calibration measurements at test points corresponding to the reference values shall be conducted after the test sets have returned to standard ambient conditions for a minimum of 12 hours. The calibration accuracy tolerances specified in 4.3.3 shall apply. No damage shall be evident which could affect subsequent operations.

4.3.7 High temperature. The test sets shall be placed within a test chamber capable of maintaining an internal temperature of 71°C, plus or minus 2°C, for a 12-hour period. At the end of the 12-hour period, the temperature shall be lowered to 55°C, plus or minus 2°C, for a 2-hour period. At the end of this period, and while the test sets are still at high temperature, they shall be examined and checked to determine that they operate satisfactorily and that the calibration accuracy does not differ from the reference values by more than plus or minus 1 percent of the reading, or plus or minus 0.5 percent of

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the maximum value of each capacitance range, whichever is greater. For the resistance section, the readings noted shall not differ from the reference values by a resistance value equal to plus or minus 10 percent of the reference values. Calibration measurements at test points corresponding to the reference values shall be conducted after the test sets have returned to standard ambient conditions or a minimum of 12 hours. The calibration accuracy tolerance specified in 4.3.3 shall apply. No damage shall be evident which could affect subsequent operations.

4.3.8 Vibration. The test sets with the lid closed and secured, shall be mounted on the vibration test stand. The test sets shall be vibrated with the frequency varying between 10 and 55 Hz at a total excursion of 0.03 inch (0.762 mm), plus or minus .006 inch (0.152 mm). The frequency shall be varied uniformly from 10 to 55 Hz and returned to 10 Hz in approximately 1 minute. The vibration shall be applied in each of the following directions for a period of 1 hour:

a. Horizontally, parallel to the major horizontal axis of the test sets.

b. Horizontally, at right angles to the major horizontal axis of the test sets.

c. Vertically.

Upon completion of the vibration, the test sets shall be inspected to ascertain that no fixed parts have become loose or damaged. The calibration accuracy of the test sets shall be checked and shall be within the applicable tolerances specified in 4.3.3.

4.3.9 Temperature cycling. The test sets shall be placed within a test chamber capable of maintaining internal temperatures of minus 55°C, plus or minus 2°C, and 70°C, plus or minus 2°C, respectively for a 2-hour period. The test sets shall be subjected to a minimum of 5 temperature cycles. Each cycle shall consist of starting at room temperature, lowering and maintaining the temperature minus 55°C, plus or minus 2°C, for a 2-hour period; returning to standard ambient conditions for a 2-hour period; raising and maintaining the temperature at 70°C, plus or minus 2°C, for a 2-hour period; and returns to room temperature for a 2-hour period. Calibration measurements at points corresponding to the reference values shall be conducted after the test sets have returned to room temperature for approximately 12 hours. The calibration accuracy tolerances specified in 4.3.3 shall apply. No damage shall be evident which could affect subsequent operation.

4.3.10 Shock. The test set shall be tested for shock in accordance with Procedures VI, Bench Handling, Method 516.3 of MIL-STD-810.
Calibration measurements at test points corresponding to the reference values shall be conducted after the test sets have been subjected to the above-mentioned shock conditions. The calibration accuracy tolerance specified in 4.3.3 shall

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apply. No damage shall be evident which could affect subsequent operation of the sets.

4.3.11 Humidity. The test sets shall be subjected to the humidity test in accordance with Procedure I of Method 507.2, MIL-STD-810. The test sets shall be mounted in a normal operating position with the lid open and cables and adapters removed. Within 1 hour after the end of the period, the calibration accuracy of the test sets shall be checked. The readings established for the capacitance section shall not differ from the reference values by more than plus or minus 3 percent of the reading, or 1.5 percent or the maximum value of each capacitance range, whichever is greater. For the resistance section, their readings shall be noted and shall not differ from the reference values by a resistance value equal to plus or minus 15 percent of the reference values. The external moisture on the panel shall be wiped off with a cloth before testing. The test sets shall be visually inspected to determine that no corrosion or other deterioration exists which could affect subsequent operation.

4.4 Inspection of preparation for delivery. Preservation and packaging, packing, and marking shall be inspected to determine compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation and Packaging. Test sets shall be preserved and packaged in accordance with level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Test sets, assembled into instrument case and transit case, shall be packaged one each in snug fitting fiberboard boxes conforming to W5c or W5s of PPP-B-636.

5.1.2 Level B. Test sets shall be packaged as for Level A, except that PPP-B -636, class domestic may be used as unit container.

5.1.3 Commercial. Test sets shall be packaged in accordance with ASTM D 3951.

5.2 Packing. Test sets, packaged as per 5.1, shall be packed Level A or B as specified (see 6.2).

5.2.1 Level A. Test sets shall be packed in snug fitting plywood containers conforming to PPP-B-601, grade A.

5.2.2 Level B. Test sets shall be packed in snug fitting fiberboard boxes conforming to V2s of PPP-B-636.

5.2.3 Commercial. Test sets shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special requirements of the contract or

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order, interior and exterior containers shall be marked in accordance with MIL-STD-129, except that commercial packaging/packing be marked in accordance with ASTM D 3951.

5.4 Commercial packaging. When specified in the contract or order, preservation, packing, and marking, as applicable, shall be acceptable if they meet the requirements of ASTM D 3951

6. NOTES

6.1 Intended use. The variable capacitance fuel-quantity-gage test set is for use in measuring the capacitance of fuel-gaging systems, both compensated and non-compensated. It is also for use in measuring insulation resistance of the fuel-gage tank units of the capacitance type, and testing and calibrating the fuel-quantity indicators. The test set is designed for field and shop use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether first article approval and/or functional tests are required (see 3.1.1).

c. Level of preservation and packaging, and packing required (see 5.1 and 5.2).

d. Whether special marking is required (see 5.3).

6.3 Definitions.

6.3.1 Transit case. A transit case is a case without a test instrument built in. The transit case is primarily a protective case intended to house the equipment and accessories. It is not to be confused with a packing case, and is not intended for initial shipment without additional packing.

6.3.2 Instrument case. An instrument case is a case protecting the instrument proper and is part of the instrument. Thus an instrument panel may be part of an instrument case.

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-58092D(AV)	Test Set, Fuel-Quantity Gage: Capacitance Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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